

UMGT Engine Control

Operating Manual & Reference

Micro gas-turbine ECU — ESP32 firmware + UMGTEngine dashboard

This manual describes how to connect, configure and operate the UMGTEngine turbine engine control system: every panel, button, parameter and serial command in the UMGTEngine application, the automatic start sequence, the failsafe logic, and the ESC / fuel / sensor subsystems.

Generated for the UMGTECU 2026 project

1. System Overview

The UMGH engine control system has two halves that talk to each other over a single USB serial link:

- **The ECU (firmware)** — an ESP32 DevKitC-32 on a custom carrier board ("ESPSheild"). It reads the sensors, drives the actuators, runs the automatic start/shutdown sequence, the RPM governor and all failsafes, and talks to the ESC over a DroneCAN bus.
- **UMGH_Engine.exe (the GUI)** — a Windows dashboard that connects to the ECU over USB, shows live telemetry (charts and gauges), and lets you command the engine, tune parameters and log data. It is a monitor and controller only — all real-time control happens on the ECU, so the engine stays safe even if the PC or USB link drops.

The GUI is modeled on the original LabVIEW turbine test-stand, extended with an automatic sequencer and failsafe supervisor that the LabVIEW did not have.

1.1 What the ECU controls and reads

Function	Hardware	Notes
ESC / starter-generator	Hargrave microDRIVE LPi over DroneCAN (CAN 1 Mbps)	Throttle, RPM, brake; reads RPM / voltage / current / temperature
Fuel pump 1	MOSFET M5 (GPIO27), Hausl ZP25M14F gear pump	PWM, ml/min via calibration curve
Fuel pump 2	MOSFET M4 (GPIO26)	Independent second fuel pump
Glow plug	MOSFET M1 (GPIO32)	On/off ignition heater
Solenoid 1	MOSFET M3 (GPIO25)	PWM to ~6 V (configurable duty)
Solenoid 2	MOSFET M2 (GPIO33)	PWM to ~6 V (configurable duty)
Thermocouples x4	MAX31855 (SPI), CS via PCF8575 I2C expander	TC1 TIT, TC2 EGT, TC3 Bearing, TC4 Coil
Current sensors x2	QNDB6 hall sensors on GPIO34 / GPIO35	Battery / load current
Pixhawk link	Serial2 (GPIO16/17)	Optional telemetry bridge

1.2 The USB data link

The ECU streams a compact **DATA:** line at up to 50 Hz (default 10 Hz) carrying every live value, plus a one-per-second **HB** heartbeat so the GUI always knows the ECU is alive even when streaming is off. Commands are plain text typed or sent as button presses.

NOTE: The GUI enables the stream automatically when you connect. If you ever see the heartbeat but no live data, type `stream` on in the terminal — the board is fine, the stream was just off.

2. Getting Started

2.1 Connecting

- 1 Plug the ESP32 into USB. Launch **UMGT_Engine.exe**.
- 2 Pick the COM port from the drop-down (top-right). If unsure, click the refresh arrow and choose the CH340 port.
- 3 Click **Connect**. The ESP32 reboots (~2 s), then the status turns green “Connected” and data begins to flow.
- 4 Within a few seconds the GUI pulls the saved parameters from the ECU into the parameter panel.

NOTE: The USB-serial port number can change between plug-ins (COM4/COM5/etc.). Just pick whichever port appears. If none appears, unplug and replug the ESP32's USB cable.

2.2 The header bar



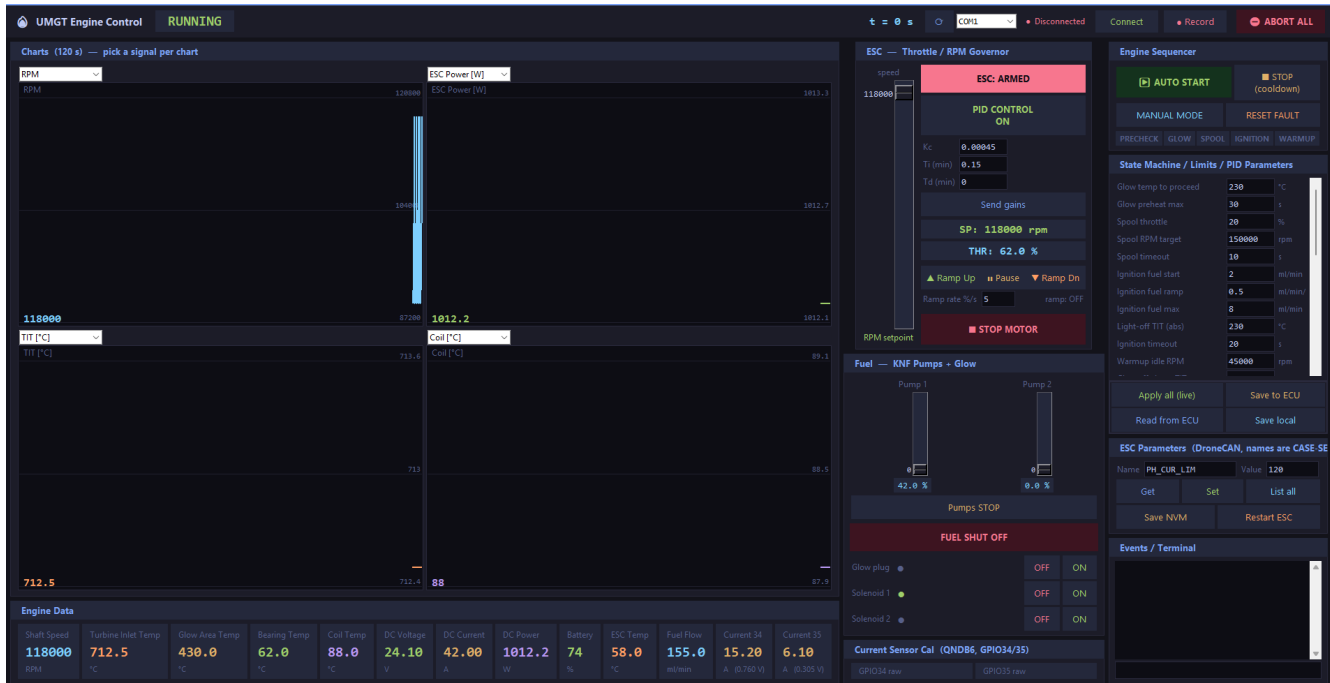
The header bar, left to right: (1) engine-state chip, (2) COM port, (3) Connect, (4) Record, (5) ABORT ALL.

Control	What it does
Engine state chip	Large label showing the current sequencer state (OFF, MANUAL, PRECHECK, GLOW, SPOOL, IGNITION, WARMUP, RUNNING, COOLDOWN, FAULT).
FAULT text	Appears in red next to the state when a failsafe has tripped, naming the fault.
t = <n> s	Seconds since connecting.
Port drop-down + refresh	Select / rescan the serial port.
Connect / Disconnect	Open or close the serial link.
Record	Start/stop CSV logging of every telemetry field to a file you choose.
ABORT ALL	Emergency stop — immediately cuts fuel, glow, throttle and disarms the ESC, latching a FAULT. Always available.

WARNING: The **ABORT ALL** button (and the `eng abort` command) is your emergency stop. It kills every actuator instantly. Know where it is before running the engine.

3. The Dashboard, Panel by Panel

The window is arranged in three columns. The left column holds the live charts and the engine-data gauges; the middle column holds the ESC/throttle and fuel controls; the right column holds the sequencer, the parameter editor, the ESC parameter tool and the event terminal.



The full UMG_Engine dashboard (shown with simulated running data). Left: charts + engine-data gauges. Middle: ESC/throttle and fuel. Right: sequencer, parameters, ESC parameters, terminal.

3.1 Charts (top-left)

Four strip charts scroll the last 120 seconds. Each chart has a **drop-down at its top-left** — pick any signal to plot: RPM, ESC Power, TIT, EGT/Glow, Bearing, Coil, DC Voltage, DC Current, ESC Temp, Battery, Current 34, Current 35, Fuel Flow, or Throttle. Defaults are RPM, ESC Power, TIT and Coil.

NOTE: A chart only draws a signal that has data. If a chart looks blank, the signal it is set to has no valid readings right now (e.g. RPM when the ESC is off, or a thermocouple that reads OPEN). Point it at a live signal.

3.2 Engine Data gauges (bottom-left)

A row of numeric gauges with warning colors: Shaft Speed (RPM), Turbine Inlet Temp, Glow Area Temp, Bearing Temp, Coil Temp, DC Voltage, DC Current, DC Power, Battery %, ESC Temp, Fuel Flow (ml/min), and the two hall Current sensors (34 & 35). A temperature gauge turns amber near its limit and red at/over it. A thermocouple with no probe shows **OPEN** in red.

The two current gauges also show the raw ADC voltage underneath (e.g. “A (1.505 V)”) to help diagnose the sensor.

3.3 ESC — Throttle / RPM Governor (middle)

- **ESC: ARMED / DISARMED button** — the microDRIVE refuses to spin the motor unless the ECU broadcasts an arming signal. Click to arm (with a confirmation); the button turns red when armed. Arm before commanding throttle; disarm (or ABORT) to make it safe.
- **PID CONTROL button** — toggles the RPM governor. Off = the big slider is manual throttle %. On = the slider is an RPM setpoint and the PID holds that RPM.
- **Kc / Ti / Td fields + Send gains** — the governor PID gains (Ti and Td in minutes, matching the LabVIEW convention). Type new values and click Send gains.
- **SP / THR readouts** — the current RPM setpoint and the throttle % the ECU is actually commanding.
- **Speed slider** — the large vertical slider. It sets throttle % (governor off) or RPM setpoint (governor on). It auto-switches labels.
- **Ramp Up / Pause / Ramp Down + Ramp rate** — ramps the throttle up or down at the configured %/s instead of jumping.
- **STOP MOTOR** — cancels ramp and governor and sets throttle to 0 (stays armed).



ESC / Throttle / RPM Governor panel.

- 1 ESC: ARMED / DISARMED** — arm the ESC (red = armed).
- 2 PID CONTROL** — toggle the RPM governor.
- 3 Send gains** — push the Kc / Ti / Td fields above it.
- 4 SP** — current RPM setpoint.
- 5 THR** — throttle % the ECU is commanding.
- 6 Speed slider** — throttle % (governor off) or RPM setpoint (governor on).
- 7 Ramp Up / Pause / Ramp Down** — ramp the throttle at the set rate.
- 8 STOP MOTOR** — throttle to 0 (stays armed).

3.4 Fuel — Pumps + Glow + Solenoids (middle)

- **Pump 1 / Pump 2 sliders** — set each fuel pump's PWM duty 0–100%. Fuel flow is shown in ml/min via the pump calibration curve.

- **Pumps STOP** — both pumps to zero.
- **FUEL SHUT OFF** — a latch that forces both pumps to 0 and blocks them until released. Turns dark red when engaged.
- **Glow plug ON / OFF** with a status lamp.
- **Solenoid 1 / Solenoid 2 ON / OFF** with lamps. “On” drives the MOSFET at the Solenoid-ON-duty parameter (default 50% ≈ 6 V on a 12 V bus).



Fuel panel: pumps, glow plug, solenoids.

- 1 Pump 1 slider** — fuel pump 1 duty (flow shown in ml/min).
- 2 Pump 2 slider** — fuel pump 2 duty.
- 3 Pumps STOP** — both pumps to zero.
- 4 FUEL SHUT OFF** — latch that forces both pumps to 0.
- 5 Glow plug ON / OFF** — with status lamp.
- 6 Solenoid 1 / 2 ON / OFF** — drives the coil at ~6 V.

3.5 Current Sensor Cal (middle)

Live raw voltage for GPIO34 and GPIO35, plus calibration controls: a **Zero V** and **A/V** field with **Apply**; **Zero now (@0A)** which captures the present voltage as the zero point; preset buttons **Uni 100A/5V** and **Bipolar ±100A/5V**; and a **Turns** field for the multi-turn trick (pass the wire through the sensor N times to multiply a small current, then click Apply N turns).

3.6 Engine Sequencer (right)

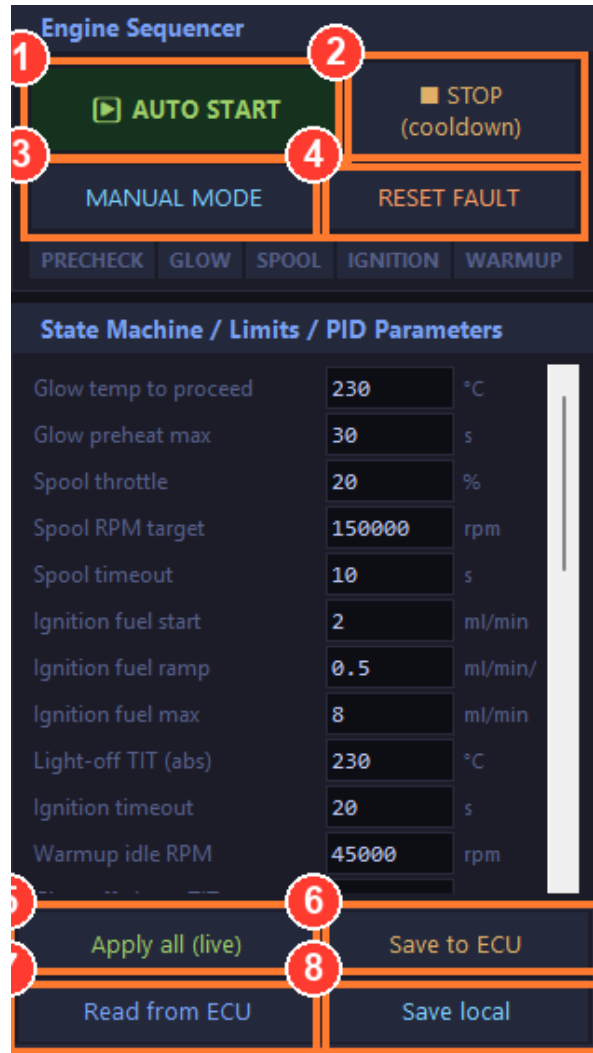
- **AUTO START** — runs the full automatic start sequence (with a confirmation dialog). See Section 4.
- **STOP (cooldown)** — normal shutdown: cuts fuel and glow, then motors the engine to cool it before stopping.
- **MANUAL MODE** — puts the ECU in manual so you can drive pumps, glow, solenoids and throttle directly. Failsafes stay active.
- **RESET FAULT** — clears a latched FAULT once the cause is resolved.

- **Sequence progress strip** — highlights the current step (PRECHECK → GLOW → SPOOL → IGNITION → WARMUP → RUNNING).

3.7 Parameters panel (right)

A scrolling list of every engine parameter with an editable field. Buttons:

- **Apply all (live)** — send all fields to the ECU's live RAM (takes effect immediately, lost on reboot).
- **Save to ECU** — apply, then persist to the ECU's flash so the values survive a power cycle.
- **Read from ECU** — pull the ECU's current values back into the fields.
- **Save local** — save the fields to a JSON file on the PC.



Engine Sequencer (top) and the Parameters panel with its four buttons.

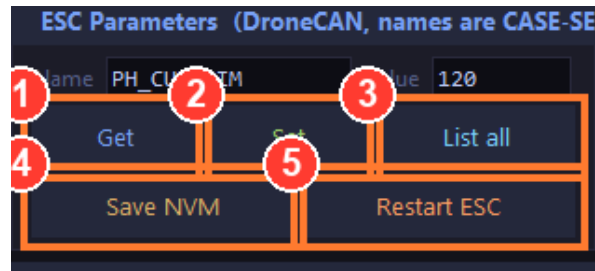
- 1 AUTO START** — run the automatic start sequence.
- 2 STOP (cooldown)** — normal shutdown with cooldown motoring.
- 3 MANUAL MODE** — direct control of each actuator.
- 4 RESET FAULT** — clear a latched fault.
- 5 Apply all (live)** — send fields to ECU RAM.
- 6 Save to ECU** — persist parameters to ECU flash.

- 7 Read from ECU** — pull the ECU's values into the fields.
- 8 Save local** — save fields to a PC file.

NOTE: Two separate memories: **Save to ECU** stores engine parameters in the ECU's flash; the **ESC Parameters** panel's **Save NVM** stores ESC settings in the ESC's own flash. They are independent.

3.8 ESC Parameters panel (right)

Reads and writes the microDRIVE's own DroneCAN settings. Type a **case-sensitive** name (e.g. PH_CUR_LIM) and a value. Buttons: **Get**, **Set**, **List all** (dumps all ~99 ESC parameters to the log), **Save NVM** (persist to the ESC — the ESC reboots itself afterward), and **Restart ESC** (reboot / clear latched ESC errors).



ESC Parameters panel (DroneCAN). Type a case-sensitive name and value.

- 1 Get** — read the named parameter from the ESC.
- 2 Set** — write the value to the ESC (live).
- 3 List all** — dump every ESC parameter to the log.
- 4 Save NVM** — persist ESC settings (ESC reboots).
- 5 Restart ESC** — reboot the ESC / clear latched errors.

3.9 Events / Terminal (right)

A scrolling log of ECU events (state changes, faults, parameter echoes) and a text box at the bottom where you can type any serial command directly (Section 7). A mid-run “ESP32 REBOOTED” line here (red if a brownout) means the board lost power and reset.

4. The Automatic Start Sequence

Pressing **AUTO START** (or `eng_start`) runs a state machine. Each step's thresholds and timeouts are parameters you can tune (Section 5). The sequence:

State	What happens	Advances when	Fails if
PRECHECK	Verifies fuel isn't shut off, TIT thermocouple is valid, ESC telemetry is present; arms the ESC and turns the glow plug on.	Checks pass	Any check fails
GLOW	Preheats the glow plug.	Glow-area temp reaches "Glow temp to proceed", or the preheat time elapses	—
SPOOL	Motors the turbine on the ESC at "Spool throttle".	RPM reaches "Spool RPM target"	RPM not reached before "Spool timeout" (SPOOL_FAIL)
IGNITION	Introduces fuel on pump 1, ramping it up in ml/min.	TIT reaches "Light-off TIT" (combustion detected)	No light-off before "Ignition timeout" (IGN_TIMEOUT)
WARMUP	Governor holds "Warmup idle RPM"; glow plug turns off once TIT passes "Glow off above TIT".	RPM is within 10% of warmup idle	—
RUNNING	Engine at idle under governor / operator control.	Operator commands	Any failsafe
COOLDOWN	On STOP: fuel and glow off, motors the engine to cool.	TIT drops below "Cooldown until TIT", or cooldown time elapses	—

CAUTION: The start parameters shipped are conservative starting points. Verify each against your engine on the bench before a real light-up — especially spool throttle vs. spool RPM target, and the ignition fuel values against what your pump can actually deliver.

4.1 Manual mode

MANUAL MODE lets you exercise each actuator independently: pump sliders, glow, solenoids, and either manual throttle or the governor. The failsafe supervisor still runs, so an over-temp or ESC-loss will still trip a FAULT and make everything safe.

5. Failsafes

Whenever anything is live (a sequence is running, or an output is on, or in manual mode), the ECU checks these every 50 ms. Any trip immediately makes everything safe (fuel, glow, throttle to zero; ESC disarmed) and latches a FAULT that names the cause. Clear it with RESET FAULT after fixing the problem. All thresholds are editable parameters.

Fault	Trips when
TIT_OVER	Turbine inlet temp reaches TIT max
COIL_OVER	Generator coil temp reaches Coil max
BEARING_OVER	Bearing temp reaches Bearing max
OVERSPEED	RPM reaches Max RPM
FLAMEOUT	Engine is lit and fuel is on but TIT falls below the Flameout TIT floor
ESC_LOSS	ESC telemetry is older than the ESC-loss timeout while throttle is commanded
TC_LOSS	TIT thermocouple goes invalid during a running sequence
IGN_TIMEOUT	No light-off within the ignition timeout
SPOOL_FAIL	Spool RPM not reached within the spool timeout
PRECHECK_FAIL	A precheck condition (fuel shutoff / TIT / ESC) not met at start
ABORT	Operator pressed ABORT ALL / eng abort

6. Parameter Reference

Every parameter is editable in the Parameters panel or with `eng set <name> <value>`. **Save to ECU** persists them to flash.

Name	Label	Unit	Meaning
glow_temp	Glow temp to proceed	°C	Glow-area temp that ends preheat and starts spool
glow_time	Glow preheat max	s	Max preheat time before proceeding anyway
spool_duty	Spool throttle	%	ESC throttle used to motor the turbine during spool
spool_rpm	Spool RPM target	rpm	RPM to reach before fuel is introduced
spool_time	Spool timeout	s	Fail to SPOOL_FAIL if not reached
ign_pump	Ignition fuel start	ml/min	Pump 1 fuel flow when fuel is first introduced
ign_ramp	Ignition fuel ramp	ml/min/s	Rate the ignition fuel is increased
ign_pump_max	Ignition fuel max	ml/min	Ceiling on ignition fuel (anti-flood)
lightoff_tit	Light-off TIT (abs)	°C	Absolute TIT that declares combustion / light-off
ign_timeout	Ignition timeout	s	Fail to IGN_TIMEOUT if no light-off
warmup_rpm	Warmup idle RPM	rpm	Governor target after light-off
glow_off_tit	Glow off above TIT	°C	TIT at which the glow plug switches off
tit_max	TIT max (fault)	°C	Over-temp failsafe on turbine inlet temp
coil_max	Coil max (fault)	°C	Over-temp failsafe on generator coil
bearing_max	Bearing max (fault)	°C	Over-temp failsafe on bearing
max_rpm	Max RPM (fault)	rpm	Overspeed failsafe + ramp/governor ceiling
flameout_tit	Flameout TIT floor	°C	Lit + fuel-on + TIT below this = flameout
esc_loss	ESC loss timeout	s	ESC telemetry age that triggers ESC_LOSS
cool_duty	Cooldown throttle	%	Motoring throttle during cooldown
cool_tit	Cooldown until TIT	°C	Cooldown ends below this temp
cool_time	Cooldown max time	s	Max cooldown duration
fuel_density	Fuel density	g/ml	Used for the g/s fuel estimate (kerosene ~0.84)
gov_kc	PID Kc	%/rpm	Governor proportional gain
gov_ti	PID Ti	min	Governor integral time (minutes)
gov_td	PID Td	min	Governor derivative time (minutes)
gov_min	PID output min	%	Lower throttle clamp for the governor (LabVIEW 15%)
gov_max	PID output max	%	Upper throttle clamp for the governor
gov_slew	Throttle slew limit	%/s	Max rate the throttle output can change
ramp_rate	Ramp rate	%/s	Manual ramp up/down rate
batt_min	Battery 0% voltage	V	Voltage mapped to 0% battery
batt_max	Battery 100% voltage	V	Voltage mapped to 100% battery
sol_duty	Solenoid ON duty	%	PWM duty for solenoid "on" (50% ≈ 6 V of 12 V)

7. Serial Command Reference

Anything the buttons do can also be typed in the Events / Terminal box. Commands are case-insensitive except ESC parameter names and serial2 send text.

7.1 Engine control

Command	Action
eng start	Begin the automatic start sequence
eng stop	Normal shutdown into COOLDOWN
eng abort	Emergency stop, latch FAULT
eng reset	Clear a latched fault
eng manual	Enter manual mode
eng status	Print full engine status
eng params	List all parameters
eng set <name> <value>	Edit a parameter (live RAM)
eng save	Persist parameters to ECU flash
eng defaults	Clear saved params (code defaults on next boot)
gov on off	Enable / disable the RPM governor
gov sp <rpm>	Set the governor RPM setpoint
gov gains <kc> <ti> <td>	Set PID gains (Ti/Td in minutes)
thr <0-100>	Manual throttle percent (governor off)
ramp up down pause off	Throttle ramp control
ramp rate <pct/s>	Set the ramp rate
pump <1 2> <0-100>	Set a fuel pump duty
pump stop	Stop both pumps
glow on off	Glow plug on/off
sol <1 2> on off	Solenoid on (drives sol_duty %) / off
fuel cut on off	Engage / release the fuel-shutoff latch

7.2 ESC / CAN (DroneCAN)

Command	Action
can	CAN status + decoded ESC telemetry
can esc	Show ESC telemetry
can arm disarm	Broadcast armed / disarmed to the ESC
can duty <-100..100>	Duty-cycle command (RawCommand)
can rpm <setpoint>	Closed-loop RPM command (RPMCommand)
can brake <0-100>	Regenerative brake (needs Reversible mode)

Command	Action
can stop	Zero the ESC command
can param list	List all ESC parameters
can param get <NAME>	Read an ESC parameter (case-sensitive)
can param set <NAME> <v>	Write an ESC parameter
can save	Persist ESC params to ESC NVM (ESC reboots)
can restart	Reboot the ESC / clear latched errors
can baud <125 250 500 1000>	Change CAN bitrate (DroneCAN = 1000)
can print on off	Toggle raw CAN frame printing

7.3 Sensors, stream & diagnostics

Command	Action
status	Full pin/sensor/health status dump
stream on off	Enable / disable the live data stream
stream rate <1-50>	Set stream rate in Hz (TCs sample at 10 Hz)
tc read <1-4 all>	Read thermocouple(s)
tc scan	Probe all expander CS bits (diagnostic)
tc mode 31855 6675	Select thermocouple chip type
current all a b	Read the hall current sensors
current cal <zeroV> <A_per_V>	Calibrate the current sensors
analog all vp vn a b	Read raw ADC inputs
i2c scan	Scan the I2C bus
i2c recover	Force I2C expander recovery
serial2 send <text>	Send text to the Pixhawk UART
serial2 baud <rate>	Set Serial2 baud (Pixhawk telem = 57600)
help ?	Print the on-board command list

8. Subsystem Notes

8.1 Fuel system & pump calibration

The fuel pump is a positive-displacement gear pump (Hausl ZP25M14F). Its flow-vs-duty curve is **nonlinear with a hard deadband**: no flow below ~28% duty, then it jumps to ~57 ml/min at 29% and rises to ~210 ml/min at 50%. The ECU stores your measured calibration points and interpolates between them, so commands in ml/min map to the correct duty and the Fuel Flow gauge is accurate.

CAUTION: Because of the deadband, this pump cannot deliver a steady flow below ~57 ml/min. If your ignition fuel needs to be lower than that, the pump must be pulsed (a firmware feature that can be added) rather than run at a low continuous duty.

8.2 ESC (Hargrave microDRIVE LPI)

The ESC speaks DroneCAN at 1 Mbps. The ECU acts as the node-ID allocator, broadcasts its own node status and the arming signal, decodes the ESC's telemetry (RPM, voltage, current, temperature, error flags), and can command duty, RPM or brake. Key ESC settings you may need (via the ESC Parameters panel): PH_CUR_LIM (phase current, 120 A for the 1500 W engine), BUS_CUR_LIM, and MOT_POLE_PAIRS (must match your motor so reported RPM is mechanical, not electrical).

WARNING: The ESC will not spin unless it is **armed** and it has seen a zero throttle first. Arm with the ESC:ARMED button (or the sequencer arms it automatically). If the ESC blinks a warning/error, use Restart ESC to clear a latched fault.

8.3 Current sensors

Two QNDB6 hall sensors (100 A / 5 V) on GPIO34 and GPIO35. Because the ESP32 ADC only reads to 3.3 V, a 5 V-output sensor clips above ~66 A on a single pass. For small currents, use the multi-turn trick in the Current Sensor Cal panel. Verify direction: a unidirectional sensor reads 0 V for reverse current — if the raw voltage sits at 0 while current flows, reverse the wire through the sensor.

8.4 Thermocouples

Four MAX31855 K-type modules on a shared SPI bus, each selected through the PCF8575 I2C expander. TC1=TIT, TC2=EGT, TC3=Bearing, TC4=Coil. A channel with no probe (or an open probe) reads **OPEN**; the chip itself still reports a healthy cold-junction temperature. The firmware auto-recovers the I2C bus if electrical noise wedges the expander.

8.5 Solenoids & glow plug

The glow plug is a simple on/off MOSFET switch (M1). The two solenoids are driven with PWM at the sol_duty parameter (default 50%), which averages through the coil inductance to ~6 V on a 12 V bus — matching a 6 V-rated coil. Tune sol_duty to hit the exact coil voltage.

9. Troubleshooting

Symptom	Likely cause & fix
Heartbeat works but no live data	The stream is off. Type <code>stream on</code> , or reconnect (the GUI enables it automatically).
Everything goes dark at once (no temp, no ESC, weak solenoid)	The ESP32 reset / browned out. Look for a red “ESP32 REBOOTED — BROWNOUT” in the log. Fix power/grounding: keep ESC and high-current returns off the ESP32 signal ground; give the ESP32 a clean 5 V. A full power-cycle recovers it.
Thermocouples all read OPEN after having worked	Usually the probes; the chips are alive. If truly no SPI data, the I2C expander wedged — <code>i2c recover</code> , or power-cycle. Check <code>status</code> for expander OK and I2C fail count.
No ESC data; can status shows <code>bus_err</code> climbing fast	ESC not on the bus. Check ESC power, CANH/CANL wiring (not swapped), and 120 Ω termination at both ends.
ESC won't spin	Not armed, or a latched ESC error. Arm it; if it still refuses, Restart ESC. Confirm <code>PH_CUR_LIM</code> and <code>MOT_POLE_PAIRS</code> .
Current sensor reads 0 while current flows	Reverse current direction (unidirectional sensor) or Common not tied to ESP32 GND. Reverse the wire through the sensor / bond the grounds.
Current reading is flickery / too low	A few amps on a 100 A sensor sits in ADC noise. Use the multi-turn trick; the firmware already low-pass filters the channel.
COM port not found / changes number	The CH340 re-enumerates on replug. Pick whichever port appears; unplug and replug USB if none shows.

9.1 Safety checklist before a run

- Know where ABORT ALL is; test that STOP MOTOR and fuel-shutoff respond.
- Confirm all failsafe limits (TIT/coil/bearing/overspeed) match your engine and are saved to the ECU.
- Verify the ESC arms and disarms, and that `MOT_POLE_PAIRS` is correct.
- Check the fuel calibration and that ignition fuel is within what the pump can deliver.
- Confirm clean, solid power to the ESP32 and proper grounding to avoid brownout resets.
- Clear the area; treat every armed/fueled state as live.

This manual reflects the UMGH Engine firmware and dashboard as configured for the project. Parameter values and pump calibration are engine-specific — always verify against your hardware before operating.